

Cell Type Bacterial, gram positive

Molecules Electroporated DNA: pGB301, 9.8 kB.

Species Used *Lactococcus lactis*, subspecies *lactis*, LM0230

Before the Pulse

Cell Growth Medium M-17 glucose (ATCC)

Growth Phase at Harvest O.D. (600) = (stationary)

Pre-pulse Incubation 1 min. on ice

Wash Solution Distilled water

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature 4 °C

Electroporation Medium Distilled water

Cuvette Gap 0.2 / 0.1 cm

Voltage 2.5 / 1.6 kV

Cell Density 5 X 10 (10) cells / ml

Volume of Cells Not given

Field Strength 12.5 / 16 kV/cm

DNA Concentration 1 µg

DNA Resuspension Buffer 10 mM Tris, 1 mM EDTA, pH 8.0

Capacitor 25 µF

Volume of DNA 3 to 5 µl

Resistor (Pulse Controller) 200 Ω

After the Pulse

Time Constant 3.0 to 3.5 msec

Outgrowth Medium M-17-Glucose (no antibiotic selection)

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

McIntyre, D.A. & Harlander, S.K.1989. *Appl. Envir. Microbiol.* **55**: 604.

McIntyre, D.A. & Harlander, S.K.1989. *Appl. Envir. Microbiol.* **55**: 2621.

McIntyre, D.A. & Harlander, S.K.1990. 'Genetic Manipulation Techniques for Lactic Cultures'. *In:Proc. XXIII Int. Dairy Cong.* **2**:1578(1990).

Outgrowth Temperature 32 °C

Length of Incubation 2 hours

Selection Method or Assay Used Erythromycin resistance

Electroporation Efficiency 10 (3) to 10 (5) transformants / µg

Per Cent Survival 10 %

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Survey Number

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